

Investigation:

Heptathlon Scoring



Baldivis
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Name: Answers

Class: _____

Teacher: _____

Marks: 29 /29

Time: 40 minutes

Equipment:

- Calculator allowed
- Pen and/or Pencil

Heptathlon Scoring

In the Olympics, “all-round” women athletes compete in the Heptathlon. Held over two days, the athletes compete in the following events:

Day 1- 100m hurdles, high jump, shot put and 200m

Day 2- long jump, javelin, and 800m

Points are awarded for each event, and the athlete with the greatest number of total points is declared the winner. But how do they decide how many points to award to a performance, and how can you compare events? For example, how do you compare a 25 second performance in the 200metre event with throwing the discus 75 metres?

Mathematics, Physics, and Computer Modelling were used by Dr Karl Ulbrich to create the current scoring system, which attempts to make fair comparisons between events.

There are three main rules used for calculating points in the seven events:

- The track events (200m, 800m and 100m hurdles): $P = a \times (b - T)^c$
- The jump events (high jump and long jump): $P = a \times (M - b)^c$
- The throwing events (shot put and javelin): $P = a \times (D - b)^c$

In these rules, P is the points scored, T is the time in seconds, M is the measurement in cm, and D is the distance in metres. The a, b and c are different for each event, as shown in the table below.

Event	a	b	c
200m	4.99	42.5	1.81
800m	0.11	254	1.88
100m Hurdles	9.23	26.7	1.84
High jump	1.85	75.0	1.35
Long jump	0.19	210	1.41
Shot put	56.0	1.50	1.05
Javelin	16.0	3.80	1.04

For example, 29.30 second performance in the 200m event would give the following number of points:

$$\begin{aligned}
 P &= 4.99 \times (42.5 - 29.3)^{1.81} \\
 &= 532.6
 \end{aligned}$$

Round Answers to 1 decimal place

1. [9 marks: 3,3 and 3]

Write the appropriate rule to work out the points in each of the following cases and then work out the points using your calculator.

a. A 32-second performance in the 200m

$$\begin{aligned} P &= a \times (b - T)^c & \checkmark \\ &= 4.99(42.5 - 32)^{1.81} & \checkmark \\ &= \underline{351.9} & \checkmark \end{aligned}$$

b. A high jump of 1.80m = 180cm

$$\begin{aligned} P &= a \times (M - b)^c & \checkmark \\ &= 1.85 \times (180 - 75)^{1.35} & \checkmark \\ &= 990.3 & \checkmark \end{aligned}$$

c. A javelin throw of 65m

$$\begin{aligned} P &= a \times (D - b)^c & \checkmark \\ &= 16 \times (65 - 3.8)^{1.04} & \checkmark \\ &= 1154.4 & \checkmark \end{aligned}$$

2. [12 marks: 5 and 7]

Using whatever method you think appropriate, find

a. The time a runner would need to score 1000 points in the 200m

$$P = a \times (b - T)^c \quad \checkmark$$

$$\frac{P}{a} = (b - T)^c \quad \checkmark$$

$$\sqrt[c]{\frac{P}{a}} = b - T \quad \checkmark$$

$$T = b - \sqrt[c]{\frac{P}{a}}$$

$$T = 42.5 - \sqrt[1.8]{\frac{1000}{4.99}} \quad \checkmark$$

$$= \underline{23.8 \text{ seconds}} \quad \checkmark$$

b. The distance in a long jump which would score the same number of points as a time of 1:55 in the 800, event

$$P = a \times (b - T)^c \quad \checkmark$$

$$= 0.11(254 - 115)^{1.88} \quad \checkmark$$

$$= \underline{1175.6}$$

$$P = a \times (M - b)^c \quad \checkmark$$

$$1175.6 = 0.19 \times (M - 210)^{1.41} \quad \checkmark$$

$$6187.368421 = (M - 210)^{1.41} \quad \checkmark$$

$$488.6651085 = M - 210 \quad \checkmark$$

$$\underline{M = 698.7 \text{ cm}} \quad \checkmark$$

3. [5 marks]

At the 1988 Seoul Olympics, just before the last event (the 800m), Jackie Joyner-Kersey of the USA needed 894 points to break the world record.

What was the slowest time which Jackie could run and still break the world record?

$$\begin{aligned}
 P &= a \times (6 - T)^c && \checkmark \\
 894 &= 0.11(254 - T)^{1.88} && \checkmark \\
 8127.2727 &= (254 - T)^{1.88} && // \\
 120.1594383 &= 254 - T \\
 \underline{T = 133.8 \text{ seconds}} &&& \checkmark
 \end{aligned}$$

*[Incidentally, Jackie ran 2:08.51, giving her a final total of 7291 points-
Still the current world record]*

4. [3 marks]

Now make up your own challenging question about the Heptathlon which can be answered using the information you have been given about the scoring system, and provide the calculations and the answer.

✓ question
 ✓ worked solution
 ✓ answer to question

